Mine Survey and Geomatics Information Management System for Land Subsidence

Mapping the Depths: Geomatics for Subsidence Management

Agenda

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Overview

Mine surveying is the process of determining the position and orientation of underground tunnels and chambers in a mine. It involves the use of specialized instruments and techniques to create maps and plans of the mine, which are used for planning, design, and safety purposes.

Geomatics Information Management Systems (GIMS) are computer-based tools that are used to manage and analyze geospatial data. GIMS can be used to store, manipulate, and visualize large amounts of data related to land subsidence, including geological, hydrological, and topographical data.

Main Causes Of Subsidence ?

Geological Process

1. Soil Compaction
2. Consolidation
3. Erosion

Human Activites

1. Extract Fulids or Solids from beneath.
2. Mining Activites
3. Urbanization

Impacts of Subsidence

1. Damage to buildings and infrastructure.
2. Increased Flooding.
3. Groundwater Depletion.
4. Damage to Ecosystems.
5. Increases Sesmic Activity.

Methods to Control or Arrest Subsidence ?

1. Artificial recharge of aquifers from the land surface.
2. Reduction of pumping draft.
3. Conservation in application and use of water : through improvement of irrigation methods.
4. Fill Subsidence Holes.
5. *__Monitor the Situation__* : - It is important to monitor the situation to ensure that the problem does not recur , This may involve ongoing surveys and data analysis.

Project objective

The main objective of this project is to device a way to develop a web-based information management system for mine survey and geomatics data to **predict land subsidence of a marked area**. The system will be designed to provide a user-friendly interface for the user to store, retrieve, and analyze various data related to mine survey and geomatics of the location.

The Process

GAMIT GLOBK

01) Data Collection

GAMIT ("GNSS at MIT") is collection of programs to process phase data to estimate three-dimensional relative positions of ground stations and satellite orbits, atmospheric zenith delays, and Earth orientation parameters. The software is designed to run under any UNIX operating system.

Implications:
We are going to use it to get data of surface height level of a given place at different points of time and use it to find the chances of land subsidence.

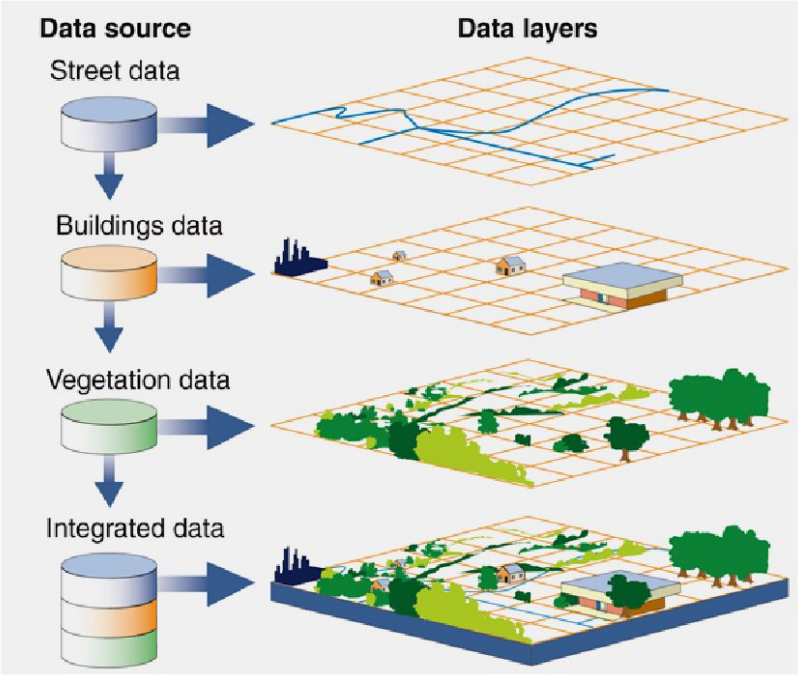
GAMIT/GLOBK

Geographical Information System (GIS)

02) Data Processing

To measure the height or coordinates of different points on the Earth's surface using satellite data, you would need to use a Geographic Information System (GIS) software that can interpret satellite imagery and provide the relevant measurements. There are several GIS software options available such as ArcGIS, QGIS, and Google Earth Pro.

Implications:
Data collected from GAMIT is processed using a GIS software, GLOBK

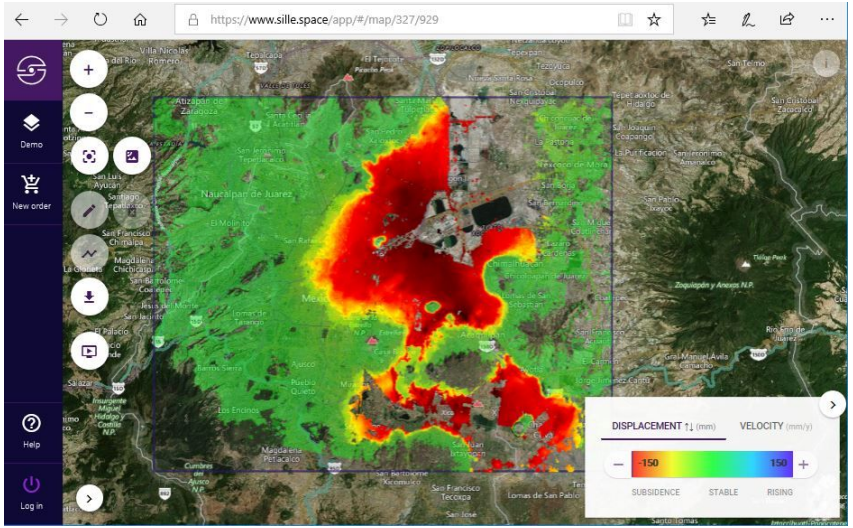


HTML

03) User Interaction via web page

HTML can be used to create the structure of the page, while JavaScript or other scripting languages can be used to add interactivity and manipulate the data. To measure the height and coordinates of a point on the map using C++, you would need to use a GIS library or SDK that provides support for satellite imagery and elevation data.

Implications:
Once we have obtained the desired measurements using GIS software, you can then create a web page to display and visualize the data.



Understanding in Depth

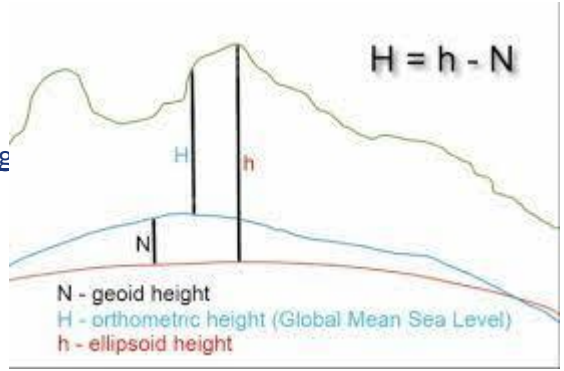
GAMIT GLOBK

For calculation of ellipsoidal height of specific point (latitude and longitude), we use GAMIT-GLOBK to retrieve information and processing.

Ellipsoidal height has been used as it provides height with respect to a fixed standard of ellipse around Earth.

The following set of commands are executed:

https://docs.google.com/document/d/1V1FY0W1FsRTqQyns4Zw2mO1zhfVk2-mg



To display Elevation data on a web page using HTML javascript and python:

A code in HTML has been made which shows the retrieved data from GAMIT GLOBK in a webpage format allowing sharing of data, easy access and user friendly view.

https://docs.google.com/document/d/1V1FY0W1FsRTqQyns4Zw2mO1zhfVk2-mgBi7ewry01Ys/edit?usp=sharing

Achievement

A web page design code is ready which shows information about the elevation data of a specified mining site.

The rate of subsidence is measured which is used for estimating the rate at future times.

These informations is used for predicting possible subsidence at future times.

(In fact as an estimated value if the rate of subsidence increases to 5 cm/year the subsidence becomes perpetuating)

All these data are presented in a user friendly format in a web page which can be easily shared and information can be searched.

Team members

Ayush Tiwari - Worked with GAMIT GLOBK software for data collection and processing of specified locations and conversion of collected data to format used for website development

Anindita Malviya - Worked on software and technologies used in GAMIT GLOBK

Dipendra Gupta - HTML website designing

Komal Arya - Research Work and Presentation

Vinit Muzalda - Land Subsidence and process of Web design

Vivek Shyamal Unadkat - Helped in deciding our mini- project, maintained the overall flow and in presentation of project.

Refrences

1. GAMIT and GLOBK at the Department of Earth, Atmospheric and Planetary Sciences, MIT.
2. GAMIT/GLOBK/TRACK | Software | UNAVCO
3. https://en.wikipedia.org/wiki/Subsidence#:~:text=In%20this%20way%2C%20land%20subsidence,taken%20into%20account%20as%20well.